Local Coverage Determination (LCD)

Facet Joint Interventions for Pain Management

L38765

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Contractor Information

Contractor Name	Contract Type	Contract Number	Jurisdiction	States
Palmetto GBA	A and B MAC	10111 - MAC A	J - J	Alabama
Palmetto GBA	A and B MAC	10112 - MAC B	J - J	Alabama
Palmetto GBA	A and B MAC	10211 - MAC A	J - J	Georgia
Palmetto GBA	A and B MAC	10212 - MAC B	J - J	Georgia
Palmetto GBA	A and B MAC	10311 - MAC A	J - J	Tennessee
Palmetto GBA	A and B MAC	10312 - MAC B	J - J	Tennessee
Palmetto GBA	A and B and HHH MAC	11201 - MAC A	J - M	South Carolina
Palmetto GBA	A and B and HHH MAC	11202 - MAC B	J - M	South Carolina
Palmetto GBA	A and B and HHH MAC	11301 - MAC A	J - M	Virginia

Contractor Name	Contract Type	Contract Number	Jurisdiction	States
Palmetto GBA	A and B and HHH MAC	11302 - MAC B	J - M	Virginia
Palmetto GBA	A and B and HHH MAC	11401 - MAC A	J - M	West Virginia
Palmetto GBA	A and B and HHH MAC	11402 - MAC B	J - M	West Virginia
Palmetto GBA	A and B and HHH MAC	11501 - MAC A	J - M	North Carolina
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Issue

Issue Description
This LCD outlines limited coverage for this service with specific details under ***Coverage Indications, Limitations and/or Medical Necessity***.

Issue - Explanation of Change Between Proposed LCD and Final LCD

Allowing medical branch block (MBB) for RFA has been added to the summary of evidence section with rationale for decision explained. The coverage was changed to allow MBB when applicable. Under Facet Cyst Aspiration/Rupture, clarification of per individual cyst was added. Under Limitations #3, added “or cyst rupture/aspiration” and clarified moderate sedation verbiage. Under Limitations #8, changed “or” to “and.”

CMS National Coverage Policy

Title XVIII of the Social Security Act, §1861(s)(2) addresses medical and other health services.

Title XVIII of the Social Security Act, §1862(a)(1)(A) allows coverage and payment for only those services that are considered to be reasonable and necessary for the diagnosis or treatment of illness or injury or to improve the functioning of a malformed body member.

Title XVIII of the Social Security Act, §1862(a)(7) excludes routine physical examinations.

Title XVIII of the Social Security Act, §1862(a)(14) defines other than physician services.

42 CFR §410.74 defines physician assistants' services, §410.75 defines nurse practitioners' services, §410.76 defines clinical nurse specialists' services, and §419.22 defines hospital services excluded from payment under the hospital outpatient prospective payment system.

CMS Internet-Only Manual, Pub. 100-02, Medicare Benefit Policy Manual, Chapter 15, §50 Drugs and Biologicals

CMS Internet-Only Manual, Pub. 100-03, Medicare National Coverage Determinations (NCD) Manual, Chapter 1, Part 1, §30.3 Acupuncture; Part 2, §150.7 Prolotherapy, Joint Sclerotherapy, and Ligamentous Injections with Sclerosing Agents; and Part 4, §220.1 Computed Tomography (CT)

CMS Internet-Only Manual, Pub. 100-08, Medicare Program Integrity Manual, Chapter 13, §13.5.4 Reasonable and Necessary Provisions in LCDs

Coverage Guidance

Coverage Indications, Limitations, and/or Medical Necessity

Facet joint interventions generally consist of 4 types of procedures: intra-articular (IA) facet joint injections, medial branch blocks (MBB), radiofrequency ablations (RFA) and facet cyst rupture/aspiration.

Facet joint interventions are considered medically reasonable and necessary for the diagnosis and treatment of chronic pain in patients who meet **ALL** the following criteria:

1. Moderate-to-severe chronic neck or low back pain, predominantly axial, that causes functional deficit measured on pain or disability scale*
2. Pain present for minimum of 3 months with documented failure to respond to non-invasive conservative management (as tolerated)
3. Absence of untreated radiculopathy or neurogenic claudication (except for radiculopathy caused by facet joint synovial cyst)
4. There is no non-facet pathology per clinical assessment or radiology studies that could explain the source of the patient's pain, including but not limited to fracture, tumor, infection, or significant deformity

*Pain assessment must be performed and documented at baseline, after each diagnostic procedure using the **same** pain scale for each assessment. A disability scale must also be obtained at baseline to be used for functional assessment (if patient qualifies for treatment).

A. **Diagnostic Facet Joint Procedures (IA or MBB):**
The primary indication of a diagnostic facet joint procedure is to diagnose whether the patient has facet syndrome. IA facet block(s) are considered reasonable and necessary as a diagnostic test only if MBBs cannot be performed due to specific documented anatomic restrictions or there is an indication to proceed with therapeutic IA injections. These restrictions must be clearly documented in the medical record and made available upon request.

Diagnostic procedures should be performed with the intent that if successful, RFA procedure would be considered the primary treatment goal at the diagnosed level(s).

A second diagnostic facet procedure is considered medically necessary to confirm validity of the initial diagnostic facet procedure when administered at the same level. The second diagnostic procedure may only be performed a minimum of 2 weeks after the initial diagnostic procedure. Clinical circumstances that necessitate an exception to the 2-week duration may be considered on an individual basis and must be clearly documented in the medical record.

a. For the first diagnostic facet joint procedure to be considered medically reasonable and necessary, the patient must meet the criteria outlined under indications for facet joint interventions.
b. A second confirmatory diagnostic facet joint procedure is considered medically reasonable and necessary in patients who meet **ALL** the following criteria:
    i. The patient meets the criteria for the first diagnostic procedure; **AND**
    ii. After the first diagnostic facet joint procedure, there must be a consistent positive response of at least 80% relief of primary (index) pain (with the duration of relief being consistent with the agent used)

Frequency Limitation: For each covered spinal region no more than 4 diagnostic joint sessions will be reimbursed per rolling 12 months, in recognition that the pain generator cannot always be identified with the initial and confirmatory diagnostic procedure.

B. **Therapeutic Facet Joint Procedures (IA or MBB):**
Therapeutic facet joint injections are considered medically reasonable and necessary for patients who meet **ALL** the following criteria:

a. The patient has had 2 medically reasonable and necessary diagnostic facet joint procedures with each 1 providing a consistent minimum of 80% relief of primary (index) pain (with the duration of relief being consistent with the agent used) **AND**
b. Subsequent therapeutic facet joint procedures at the same anatomic site results in at least consistent 50% pain relief for at least 3 months from the prior therapeutic procedure **or** at least 50% consistent improvement in the ability to perform previously painful movements and activities of daily living (ADLs) as compared to baseline measurement using the same scale; **AND**
c. Documentation of why the patient is not a candidate for RFA (such as established spinal pseudarthrosis, implanted electrical device)

Frequency Limitation: For each covered spinal region no more than 4 therapeutic facet joint sessions will be reimbursed per rolling 12 months.

C. **Facet Joint Denervation:**

The thermal radiofrequency destruction of cervical, thoracic, or lumbar paravertebral facet joint (medial branch (MB)) nerves are considered medically reasonable and necessary for patients who meet the following criteria:

a. Initial thermal radiofrequency destruction after the patient has had at least 2 medically reasonable and necessary diagnostic MBBs, with each 1 providing a consistent minimum of 80% sustained relief of primary (index) pain (with the duration of relief being consistent with the agent used)
b. Repeat thermal facet joint RFA at the same anatomic site is considered medically reasonable and necessary provided the patient had a minimum of consistent 50% improvement in pain for at least 6 months **or** at least 50% consistent improvement in the ability to perform previously painful movements and ADLs as compared to baseline measurement using the same scale.

Frequency Limitation: For each covered spinal region no more than 2 radiofrequency sessions will be reimbursed per rolling 12 months.

D. **Facet Cyst Aspiration/Rupture:**

IA facet joint injection performed with synovial cyst aspiration is considered medically necessary when both of the following criteria are met:

1. Advanced diagnostic imaging study (e.g., magnetic resonance imaging (MRI)/ computed tomography (CT)/myelogram) confirms compression or displacement of the corresponding nerve root by a facet joint synovial cyst; **AND**
2. Clinical and physical symptoms related to synovial facet cyst are documented.

Frequency Limitation: Cyst aspiration/rupture may be repeated **once** per individual cyst and only if there is 50% or more consistent improvement in pain for at least 3 months.

Limitations

1. Facet joint interventions done without CT or fluoroscopic guidance are considered not reasonable and necessary. This includes facet joint interventions done without any guidance, performed under ultrasound guidance, or with MRI.
2. Use of moderate or deep sedation, general anesthesia, and monitored anesthesia care (MAC) is not considered medically reasonable and necessary during facet procedures of IA, MBB, and facet cyst aspiration/rupture.[1]
3. The use of moderate sedation for RFA or cyst rupture/aspiration will be considered in individual cases with documentation of medical necessity such as a longstanding well-documented history of inability to cooperate, medical conditions that would prohibit performance of the procedure, or inability to remain motionless. Patient anxiety or preference alone is not sufficient justification. Routine use of moderate sedation or MAC or use of general anesthesia or deep sedation for RFA is not considered reasonable and necessary.[1]
4. It is not expected that patients will routinely present with pain in both cervical/thoracic and lumbar spinal regions. Therefore, facet joint interventions (both diagnostic and therapeutic) are limited to 1 spinal region per session.
5. It is not routinely necessary for multiple blocks (e.g., epidural injections, sympathetic blocks, trigger point injections, etc.) to be provided to a patient on the same day as facet joint procedures. Multiple blocks on the same day could lead to improper or lack of diagnosis. If performed, the medical necessity of each injection (at the same or at different level[s]) must be clearly documented in the medical record. For example, the performance of both paravertebral facet joint procedure(s) and a transforaminal epidural steroid injection (TFESI) at the same or close spinal level at the same encounter would not be expected unless a synovial cyst is compressing the nerve root. In this situation, TFESI may provide relief for the radicular pain, while the facet cyst rupture allows nerve root decompression. Frequent reporting of multiple blocks on the same day may trigger a focused medical review.
6. Facet joint IA injections and MBBs may involve the use of anesthetic, corticosteroids, anti-inflammatories and/or contrast agents, and does not include injections of biologicals or other substances not Food and Drug Administration (FDA) designated for this use.
7. One to 2 levels, either unilateral or bilateral, are allowed per session per spine region. Three or 4-level procedures are not medically necessary and therefore are non-covered. A session is a time period, which includes all procedures (i.e., MBBs, IA injections, facet cyst ruptures, and RFAs) that are performed during the same day.
8. If there is an extended time, 2 years or more, since the last RFA and there is a question as to the source of the recurrent pain then diagnostic procedures must be repeated.
9. Therapeutic facet injections are not covered unless there is justification in the medical documentation on why RFA cannot be performed. Facet joint procedures in patients for the indication of generalized pain conditions (such as fibromyalgia) or chronic centralized pain syndromes are considered not reasonable and necessary. Individual consideration may be considered under unique circumstances and with sufficient documentation of medical necessity on appeal.

10. In patients with implanted electrical devices, providers must follow manufacturer instructions and extra planning as indicated to ensure safety of procedure.

The following are considered not reasonable and necessary and therefore, will be denied:

1. IA and extra-articular facet joint prolotherapy
2. Non-thermal modalities for facet joint denervation including chemical, low-grade thermal energy (less than 80 degrees Celsius), laser neurolysis, and cryoablation
3. Intra-facet implants
4. Facet joint procedure performed after anterior lumbar interbody fusion (ALIF).
5. Definitive clinical and/or imaging findings pointing to a specific diagnosis other than facet joint syndrome
6. Diagnostic injections or MBB at the same level as the previously successful RFA procedure

Note: The scales used for measurement of pain and/or disability must be documented in the medical record. Acceptable scales include but are not limited to: verbal rating scales, Numerical Rating Scale (NRS) and Visual Analog Scale (VAS) for pain assessment, and Pain Disability Assessment Scale (PDAS), Oswestry Disability Index (ODI), Oswestry Low Back Pain Disability Questionnaire (OSW), Quebec Back Pain Disability Scale (QBPDS), Roland Morris Pain Scale (RDQ), Back Pain Functional Scale (BPFS), and the Patient-Reported Outcomes Measurement Information System (PROMIS®) profile domains to assess function.

Notice: Services performed for any given diagnosis must meet all the indications and limitations stated in this policy, the general requirements for medical necessity as stated in CMS payment policy manuals, all existing CMS national coverage determinations (NCDs), and all Medicare payment rules.

Provider Qualifications:

Medicare Program Integrity Manual states services will be considered medically reasonable and necessary only if performed by appropriately trained providers.

Patient safety and quality of care mandate that healthcare professionals who perform facet injections/procedures are appropriately trained and/or credentialed by a formal residency/fellowship program and/or are certified by either an accredited and nationally recognized organization or by a post-graduate training course accredited by an established national accrediting body or accredited professional training program. If the practitioner works in a hospital facility at any time and/or is credentialed by a hospital for any procedure, the practitioner must be credentialed to perform the same procedure in the outpatient setting. At a minimum, training must cover and develop an understanding of anatomy and drug pharmacodynamics and kinetics as well as proficiency in diagnosis and management of disease, the technical performance of the procedure, and utilization of the required associated imaging modalities.

In addition to the above requirements, non-physician providers, such as certified nurse anesthetists, with certain exceptions, may certify, order and establish the plan of care as authorized by state law. (See Sections 1861[s][2] and 1862[a][14] of Title XVIII of the Social

Security Act; 42 CFR, Sections 410.74, 410.75, 410.76 and 419.22; 58 FR 18543, April 7, 2000.) Each practitioner must provide only those services within the scope of practice for each state.

Definitions

Acute pain - The temporal definition of pain persisting for up to 4 weeks after the onset of the pain.

Axial - Relating to or situated in the central part of the body, in the head and trunk as distinguished from the limbs, e.g., axial skeleton.

Biopsychosocial model - Interdisciplinary model that looks at the interconnection between biology, pathology and socioenvironmental factors.

Central neuropathic pain - Pain, which is causally related to a lesion or disease of the central somatosensory nerves.

Centralized pain – A neurological chronic pain syndrome of the central nervous system (brain, brainstem, and spinal cord) which commonly presents with widespread generalized allodynia which is causally related to the increased responsiveness of nociceptive nerves in the central nervous system to the normal threshold or subthreshold simulation from the afferent nerves. The condition has also been called “central sensitization,” “central amplification,” and “central pain syndrome.” Fibromyalgia is considered one of the most common centralized pain syndromes.

Cervical facet pain - Pain located in the cervical spine, which may be characterized by chronic headaches, restricted motion, and axial neck pain, which may radiate sub-occipitally to the shoulders or mid-back.

Chronic pain - The temporal definition of pain persisting for greater than or equal to 12 weeks after the onset of the pain.

Dual diagnostic blocks - The diagnostic technique of injecting the same spinal nerve on 2 separate occasions to be used as an efficacy comparison to increase diagnostic accuracy.

Epidural steroid injection - The administration via injection of steroid medicine into the potential epidural space in the spinal column to deliver steroids to the spinal nerves.

Facet joint intra-articular injections, diagnostic - The placement of local anesthetic and possibly a corticosteroid into the facet joint to diagnose facet joint pain.

Facet joint intra-articular injections, therapeutic - The placement of local anesthetic and possibly a corticosteroid into the facet joint to produce the beneficial effect of pain reduction.

Facet joint - A diarthrodial joint in the spinal column (also called the zygapophysial joint or z-joint), producing the articulation of the posterior elements of 1 vertebra with its neighboring vertebra. They are bilateral superior and inferior articular surfaces at each spinal level. The

terminology or nomenclature of the facet joint is classified by the specific vertebrae level that forms it (e.g., C4-5 or L2-3). There are 2 facet joints, right and left, at each spinal level.

Facet injection (also called facet block) - A general term used to describe the injection of local anesthetic and possibly a corticosteroid in the facet joint capsule or along the MB nerves supplying the facet joints.

Facet joint denervation or radiofrequency ablation (RFA) - A general term used to describe the minimally invasive procedure that uses thermal energy generated by the radiofrequency current to deprive the facet joint of its nerve supply. The procedure is also known as a MB radiofrequency neurotomy (ablation) because it is used to thermally remove the MB nerve by using electrical current to create thermal energy to coagulate the adjacent tissues around the targeted MB nerve.

Facet joint syndrome - A set of concurrent signs or symptoms to describe facet joint pain as the pain generator. The typical clinical signs or symptoms of a facet syndrome may include local paraspinal tenderness; pain that is brought about or increased on hyperextension, rotation, and lateral bending; low back stiffness; absence of neurologic deficit; absence of root tension signs (non-radiating below the knee, absence of paresthesia). Cervical facet pain is often characterized by chronic headaches, restricted motion, and axial neck pain, which may radiate sub-occipitally to the shoulders or mid-back.

Facet level - Refers to the zygapophyseal joint or the 2 MB nerves that innervate that zygapophyseal joint. Each level has a pair of facet joints: 1 on the right side and 1 on the left side of the spine.

Intra-articular injection (IA) - The injection of local anesthetic and possibly a corticosteroid into the facet joint capsule.

Medial branch (MB) - The dorsal ramus is the dorsal branch of a spinal nerve that forms from the dorsal root of the nerve after it emerges from the spinal cord.

Medial branch block (MBB) – The placement of local anesthetic and possibly a corticosteroid near the MB nerve which supplies the sensory innervation to a specific facet joint.

Neuropathic pain – The pain which is caused by a lesion or disease of the somatosensory nerves.

Neurogenic claudication - Intermittent leg pain from impingement of the nerves emanating from the spinal cord (also called pseuduoclaudication).

New onset of spinal pain - The new onset of the spinal pain must be materially and significantly different in location, type, duration and character from the previously treated spine pain.

Non-invasive conservative management - The use of nonsteroidal anti-inflammatory drugs (NSAIDs), acetaminophen, physical therapy, acupuncture (applies to only chronic low back pain), or spinal manipulation. This management should include the application of biopsychosocial treatment techniques.

Non-radicular back pain - The radiating non-neuropathic pain which is not causally related to a spinal nerve root irritation and does not produce reproducible neuropathic symptoms in an objective dermatomal pattern.

Peripheral neuropathic pain - Pain, which is causally related to a lesion or disease of the peripheral somatosensory nerves.

Radicular back pain - The radiating neuropathic pain causally related to the spinal nerve root irritation which extends into the distal distribution, typically the lower extremity, producing neuropathic pain in a dermatomal pattern.

Radiculopathy - Radiating neuropathic pain causally related to the spinal nerve root irritation, which extends distal producing neuropathic pain in a dermatomal pattern.

Region - The segments of the back involved will be defined in this policy as 2 regions:

1. Cervical/Thoracic region= C1-C7/T1-T12
2. Lumbar/Sacral region= L1-L5/S1-S5

Session - A time period, which includes all procedures (i.e., MBBs, IA injections, facet cyst ruptures, and RFAs) performed during 1 day.

Subacute pain - The temporal definition of pain occurring during the 4-12-week time period.

Transforaminal epidural steroid injection (TFESI) - An epidural injection performed via a paramedian approach to enter the epidural space by placing the needle in the posterior-superior quadrant of the intervertebral foramen (neuroforamen) to inject near the dorsal root ganglion and exiting spinal nerve root (previously known as a selective nerve root block).

Zygapophysial joint or z-joint - See Facet joint.

History/Background and/or General Information

The spine is the most common source of chronic pain. Chronic axial spinal pain is one of the major causes of disability and accounts for a substantial United States (U.S.) health burden. Chronic spine pain poses a peculiar diagnostic and therapeutic challenge due to multiple pain sources, overlapping clinical features, and nonspecific radiological findings.

The facet joints can cause axial spinal pain and referred pain in the extremities. The pathology of the pain source is due to facet joints being richly innervated by the nerve fibers from the MB of the dorsal ramus of spinal nerves. Each facet has a dual nerve supply. One exception is at the C2–C3 zygapophysial joint, which has a singular nerve supply from the third occipital nerve (the superficial MB of C3 dorsal ramus).[2]

Facet joint interventions may be used in pain management for chronic cervical/thoracic and back pain arising from the paravertebral facet joints. The facet block procedure is an injection of a local anesthetic, with or without a steroid medication, either into the facet joint (IA) or outside the joint space around the nerve supply to the joint (the MB nerve) known as medial branch block (MBB). Imaging guidance (fluoroscopy or CT per code descriptor) is used to assure accurate placement of the needle for the injection. Paravertebral facet joint denervation is a therapeutic intervention used to provide both long-term pain relief and reduce the likelihood of recurrence of chronic cervical/thoracic or back pain confirmed as originating in the facet joint's MB nerve.[2]

There are various methods that may be used in performing facet joint denervation. Percutaneous RFA is a minimally invasive procedure done with imaging guidance (fluoroscopy or CT per code descriptor) and involves using energy in the radiofrequency range to cause necrosis of specific nerves (MBs of the dorsal rami), preventing the neural transmission of pain. Conventional RFA (non-pulsed or continuous) applies thermal energy of typically 80 to 85 degrees Celsius. The terms RFA and radiofrequency neurotomy are used interchangeably. Both terms refer to a procedure that destroys the functionality of the nerve using radiofrequency energy. Non-thermal methods of denervation include chemical (chemodenervation), low-grade thermal energy (less than 80 degrees Celsius), pulsed RFA, laser neurolysis, and cryoablation.[2]

Throughout this document, societal recommendations with the grading of evidence are referenced. There are multiple systems to grade or rank the quality of medical evidence and develop evidence-based recommendations. Not all grading systems are equivalent, so while there are typically similarities in the grades or recommendations from various grading systems, they must be considered independent of the other. The references in this document refer to the following grading systems:

1. Grades of Recommendation, Assessment, Development, and Evaluation (GRADE) Guidelines used in some systematic reviews, the basis for North American Spine Society (NASS) recommendations aligns with GRADE.

2. A modified approach to the grading of evidence[3] and development of interventional pain management specific instrument[4] used in American Society of Interventional Pain Physicians (ASIPP) Guidelines and some systematic reviews.

3. U.S. Preventive Services Task Force grading of evidence guidelines used by 2020 Consensus Guidelines by Cohen, et al.[5]

4. Levels of Evidence for Primary Research Question and Grades of Recommendation for Summaries or Review of Studies adopted by NASS.[6]

A multi-MAC Subject Matter Expert (SME) panel on facet joint and medial nerve branch procedures meeting was held on 5/28/2020.

Summary of Evidence

Diagnostic Facet Joint Injections

Due to the lack of reliable history, physical exam, or imaging to predict response, providers must rely on facet interventions diagnostic injections given for diagnostic purposes to determine if the facet joint is the source of suspected spinal pain. There is controversy over optimal patient selection for diagnostic injections, which measures successful response and type and number of diagnostic injections performed.

Numerous investigations have been undertaken to correlate symptoms and physical exam findings with facet pathology and have concluded conventional clinical findings are unreliable in identifying facet joint success.[5,7,8] A 2020 summary of the literature by Cohen, et al. “Consensus practice guidelines on interventions for lumbar facet joint pain from a multispecialty, international working group” (2020 Consensus Guidelines) included 21 studies evaluating the association of physical exam findings with facet block results, and conclude there was no historical or physical exam findings that could reliably predict response to facet joint blocks, grade C evidence, low level of certainty.[5] This is consistent with ASIPP 2020 “Comprehensive Evidence-Based Guidelines for Facet Joint Interventions in the Management of Chronic Spinal Pain: American Society of Interventional Pain Physicians (ASIPP) Guidelines" (2020 ASIPP Guidelines) which rated accurate diagnosis of facet joint pain with physical examination and symptoms level IV with a weak strength of recommendation.[8] 2020 NASS Guidelines reported insufficient evidence to make a recommendation for or against patient-reported reproduction of pain during a facet joint injection as a predictor of response to dual diagnostic blocks with a grade 1 recommendation.[6,8] A careful medical history and exam remain an important component to the evaluation of other etiologies of low back pain and raise suspicion of facet source.[8] Physical findings can help identify levels for blocks, and maneuvers can aid in the detection of radicular symptoms.[5]

Imaging studies have been investigated as a marker of painful lumbar facet joints. While degenerative changes can be found in imaging, they have not been correlated as a reliable predictor of success with facet interventions.[5,8] Nonetheless, imaging studies often play an important role in the exclusion of other etiologies of back and cervical/thoracic pain. Single-photon emission computed tomography (SPECT), a nuclear medicine imaging technique, can identify active inflammatory markers as seen in facet disease but have not shown consistent results in the prediction of facet joint intervention success. 2020 Consensus Guidelines reported on 12 studies and concluded moderate evidence for SPECT (grade C, moderate certainty) and weak evidence for scintigraphy, MRI, and CT with no to weak supporting evidence.[5] 2020 ASIPP Guidelines reported level III evidence for SPECT with weak strength

of recommendation and level V evidence for other imaging modalities, weak recommendation.[8]

Conservative treatment, also called medical/interventional treatment, includes integrative treatments (such as acupuncture (see NCD 30.3) or spinal manipulation); physical treatments (physical therapy including exercise, heat and cold modalities, massage), medications (NSAIDS/antidepressants), and lifestyle modifications (weight loss, sleep hygiene, nutrition, smoking cessation) are typically perused before interventional procedures. Facet joint pain is frequently accompanied by other pain etiologies, which may benefit from conservative modalities or combined therapy, and it is well accepted that some back pain will resolve spontaneously or with conservative or non-invasive measures over time. A few studies have had a conservative arm, or investigated combined modalities that support a combined approach, but did not find a conservative approach alone consistently lead to improvement.[6,9] The Agency for Healthcare Research released a systematic review of non-invasive, non-pharmacological treatment for chronic pain, including neck and low back. Two hundred two trials were included which evaluated the spectrum of non-invasive treatment modalities, concluded durable slight to moderate improvement in function and pain for specific chronic pain conditions and support clinical strategies that focus on non-pharmacological and non-invasive therapies.[10] 2015 NASS Guidelines recommend a failure of at least 4 weeks of non-invasive care[11]; 2020 Consensus Guidelines recommend a 3-month trial of different non-invasive treatment[5]; 2020 ASIPP Guidelines rated level of evidence II with a strong recommendation for 3 months after onset and failure of conservative management[8]; European Guidelines extend the period to 1 year.[12] Chronic pain may lead to stress, anxiety, poor sleep and depression, and other psychological sequalae and treatment beyond the pain symptoms may be necessary. Optimally a biopsychosocial model that promotes a multi-disciplinary team approach addressing physical pain and psychological factors are utilized.[13] Patients with chronic pain may also benefit from cognitive behavioral therapy, biofeedback, and other psychological interventions.[40]

Due to the lack of reliable history, physical exam, or imaging to predict response to facet interventions, diagnostic injections may be given for diagnostic purposes to determine if the facet joint is the source of suspected spinal pain. The response pattern to diagnostic facet joint injections has become the gold standard for diagnosing facet syndrome. Temporary or prolonged abolition of the spinal pain suggests that facet joints were the source of the symptoms.[2] The possibility of false-positive and false-negative reporting is an inherent risk with facet blocks and challenges identifying the patients most likely to have a positive response to treatment.[5,8] IA and MBBs have been evaluated for predictive value for successful RFA. 2020 ASIPP Guidelines reported on 10 studies in the lumbar spine and found prevalence rates ranged from 27% to 40% with false-positive rates of 27% to 47%, with ≥ 80% pain relief. They conclude level 1-2 evidence with moderate to strong strength of recommendation of lumbar, diagnostic facet nerve blocks.[8] While both IA and MBBs have been shown in studies to be predictive of success with RFA (2020 Consensus Guidelines grade B recommendation, low level of certainty)[5] 5 studies comparing lumbar MBB to IA injections, including 2 randomized studies[14,15] supports MBB as the preferred modality,

largely due to high technical failure rate and pain associated with the IA injections. One randomized trial focusing on younger patients shows IA injection with steroids may play a role in patients in which MBB is contraindicated (grade C evidence, moderate certainty).[5] The Greater Manchester EUR Policy Statement on Facet Injections no longer commissioned IA facet injections and recommended MB nerve blocks for diagnostic evaluation for RFA.[12] The Spinal Intervention Society (SIS) recommended that MBBs replace IA injections as a diagnostic indicator.[2]

In order to determine if a block is successful, there must be an assessment/measurement of pain and function. There are 3 points in which pain/function must be evaluated: 1. baseline; 2. after a diagnostic block; and 3. at each follow-up to evaluate long term relief. The definition of success lacks consistency, and there are different cut-off values in the literature. Among the multi-MAC SME panel, there was an agreement that subjective improvement in pain is a valid measurement of improvement in pain (average voting score of 3.8/5). However, there was not agreement on what the cut-off should be, if a tool should be used, and if so, which tool. There was also a consensus that function is the most important indicator of success. A study assessed the validity of subjective rating improvement after 1 IA joint injection and compared subjective rating to use of VAS scoring and concluded that validity of pain provocation alone as criterion standards in patients undergoing diagnostic facet injections should be questions with the positive predictive value of 16% for predicating facet joint as a source of pain.[16] Clinical trials and policy have used multiple standardized tools to measure pain, including the NRS and VAS, with pre-set cut-off values in attempt to determine a specific measurement in which patient's may most benefit from intervention. Studies have evaluated if a specific cut-off value can predict outcomes of RFA and do report a high correlation between verbal rating scales, NRS and VAS score in prediction of outcomes with RFA. However, the panel did not agree on a minimum value that can be used with these tools. The panel did agree that measurement of function can provide valuable clinical input into improvement, such as the ability to stand, walk, and the ability to do ADLs and can predict success with future therapeutic procedures. There are multiple tools to measure function including the PDAS[17], ODI[18], OSW, QBPDS[19], RDQ[20], BPFS[21], and the easy-to-use PROMIS® profile domains which have been found to correlate well with the ODI scale.[22]

For the diagnostic block, the pain relief achieved is temporary and used as a predictor of success for subsequent RFA. The objective is to reduce false-positives that would not be predictive of success with RFA and avoid false-negative patients who could benefit but would not be offered treatment. Most studies have used a cut-off of pain relief greater than 80% to consider MBB as positive. Ten studies assessing the prevalence of lumbar facet joint pain using a cut-off value of 80% relief reported with a prevalence rate of 27% to 40% with false-positive rates of 27% to 47% receiving a moderate to strong recommendation from 2020 ASIPP Guidelines.[8] One paper reports dual comparative blocks are advocated as a means of identifying true-positive cases and excluding placebo responders and have been shown to have a sensitivity of 100% and a specificity of 65%.[23]

As stated in the 2020 Consensus Guideline: “the cut-off designating an MBB as positive is one of the most controversial areas in pain medicine.”[5] Cohen, et al. in 2013, prospective study to evaluate predictive values and reported no difference in the predictive value of ≥ 50% to <80% pain reduction vs. ≥80% pain reduction. The guidelines advocate for using ≥50% for clinical trials and clinical practice. The guidelines acknowledge that the existing evidence does not adequately address the 50-80% group and that this cut-off was selected to maximize access to care given the lack of reliable alternative treatment options in this population and potential benefit of treatment in this group.[5] On the contrary, there are studies that show patients with ≥80% relief are more likely to show a positive response to RFA. 2020 ASPP Guidelines report level 1 to II evidence based on 10 diagnostic accuracy studies (using ≥75% and ≥80% criterion) and offer moderate to strong strength of recommendation. They cite the literature using the 50% cut-off is conflicting due to internal inconsistencies.[8]

There is controversy regarding the number of diagnostic blocks needed to ensure an accurate diagnosis before proceeding with RFA. Derby, et al. evaluated the correlation of lumbar MBB with diagnostic MBB cut-off values to optimize therapeutic outcomes and a failure of single MBB and conclude with a dual block protocol a 70% cut-off value was acceptable. With a single block, 80% or higher was optimal.[24,25] Three systematic reviews also showed significantly better improvement with dual MBB with 80% cut-off value.[26-28] Two randomized controlled trials (RCTs) (1 in the cervical region) demonstrated positive results with dual blocks.[29,30] SIS and ASIPP advocate for dual diagnostic blocks, and diagnostic accuracy is reduced if lower cut-off criteria are used. In contrast, the 2020 Consensus Guidelines advocate for single blocks using a 50% cut-off. One RCT compared outcomes of stratified by different prognostic values and reported the success rate in those who underwent RFA in the zero (no block), single and double block groups were 33%, 39%, and 64%, respectively.[31] Another study reporting on success rates of RFA based on the number of blocks concludes no difference in outcomes based on the number of blocks. A meta-analysis of 5 RCTs using zero or 1 block favored RFA.[46] However, a meta-analysis is challenging given the high heterogenicity of the literature. The 2020 Consensus Guidelines suggest the number of blocks depends on the goals with dual blocks preferable for research where diagnostic accuracy is critical while single (or no) blocks reduce the number of procedures and access to care. Balancing these factors, they recommend a single block reporting moderate evidence that dual blocks result in higher success with RFA, but zero-block results in the highest number of patients with a positive response to RFA.[5] The result of voting from the multi-MAC SME panel was a score of 3.2/5, stating confidence in the clinical literature to support 2 MBBs needed to diagnose facet pain.[11] NASS Guidelines recommend dual MBBs on 2 separate occasions and second block administered only if >80% relief with the first block due to an unacceptably high false-positive rate of single diagnostic injection. In the 2020 NASS Evidence-Based Clinical Guidelines for Multidisciplinary Spine Care, they state a single diagnostic IA injection with 50% pain relief provides no clinically meaningful improvement at 6 months (grade of recommendation B) and insufficient evidence to make a recommendation for or against the use of RFA (grade of recommendation I [Insufficient]).[6]

The literature and societal guidance report that diagnostic injections should be MBBs, not IA injections. This is agreed upon in the 2020 Consensus Guidelines, 2020 ASIPP Guidelines, and NASS Guidelines. NASS Guidelines state IA facet injections have not been validated for diagnostic use, and the false-positive rate is unknown, as well as lack of effectiveness studies comparing IA vs. MBB in the cervical spine.[11] The exceptions in the case of the atlanto-occipital and atlanto-axial joints, since there are no MB or other intervention to block readily. In these cases, dual blocks with 80% relief are supported by most guidelines.[6]

Therapeutic Joint Injections

IA injections into the facet joint have been used as a treatment modality for facet related pain and remains controversial. Therapeutic injections typically include a corticosteroid as there would not be anticipated long-acting pain relief from a short-term local anesthetic. Among the multi-MAC SME panel, the average voting was 2.70, with a range from 1-4. There is clinical evidence against the use of IA therapeutic facet joint injections. Lilius, et al. conducted an RCT including 109 patients who failed to show a difference between saline, steroids, and anesthetic injected around 2 facet joints.[32] Carette, et al. conducted an RCT, including 101 patients with IA lumbar facet pain, that showed a non-significant difference between the injection of saline and depo-corticosteroid with 22% in steroid group compared to 5% in saline group reporting benefit at 6 months.[33] A small RCT, including 41 patients, reported return of pain in 3 days after IA steroid injection compared to 3.5 days in the control.[34] Kennedy, et al. performed a small RCT comparing IA facet injections with a steroid to saline and did not find a change in need for RFA.[35] In the Facet Treatment Study (FACTS), a double-blinded RCT, Cohen, et al. found no significant difference for pain relief or functional outcome change between lumbar facet IA injection with steroid or anesthetic or MBB with steroid, anesthetic or saline for up to 6 months post-injection. Five patients made it to the 6-month follow-up, of which 2 had RFA. The FACTS study also found that MBB for anesthetic or steroids did not perform better than saline.[12]

On the contrary, a systematic review by Manchikanti, et al. (2015)[26] using a modified approach to the grading of evidence, reported 20 RCTs assessed with moderate to high-quality methodological criteria. For therapeutic interventions, the evidence was variable from level II to III, with level II evidence for lumbar facet joint nerve blocks and radiofrequency neurotomy for long-term improvement (greater than 6 months), and level III evidence for lumbosacral zygapophysial joint injections for short-term improvement only. This review provides evidence for the diagnostic validity of facet joint nerve blocks, and moderate evidence for therapeutic radiofrequency neurotomy and therapeutic facet joint nerve blocks in managing chronic low back pain.[7] These 2 studies have been criticized for a high proportion of patients taking opioids and prior back surgery, lack of control groups, and the absence of blinding.[5] Schneider, et al. reported in rigorously selected patients, 56% of patients had 100% pain relief at 6 months.[36] A systematic review by Engel reported sustained relief of 6 months, and 1/3 had relief at 1 year.[27]

This is in contrast with another systematic review during the same period that looked at the evidence supporting the use of therapeutic IA facet joint injections for patients with suspected facet joint pain. The review focused on IA facet joint injections with active drug or placebo/inactive injection are more effective in reducing back pain and back pain-related disability than conservative treatment. A total of 391 records were screened, and 6 trials were included. The trials included were small (range 18-109 participants), and overall, in terms of pain and disability outcomes, most were inconclusive. Only 2 of the trials report any significance between the group's differences in pain or disability outcomes. The authors addressed the limitations and flaws in these trials that were clinically diverse and precluded any meta-analysis. Several methodological issues were identified. The positive results are interpreted with caution and suggest that there is a need for further high-quality work in this area.[37]

Many societal recommendations that do not support the use of therapeutic facet joint injections include:

- The 2020 Consensus Guidelines developed by a multispecialty international working group recommend against the routine use of therapeutic facet injections giving it a grade D, moderate level of certainty.[5]
- National Institute for Health and Care Excellence (NICE) guideline on low back pain and sciatica in over 16s: assessment and management[38] recommendations are based on systematic reviews of best available evidence and explicit consideration of cost effectiveness. The NICE recommendation for image-guided facet joint injections “do not offer spinal injections for managing low back pain. The Guideline Development Group (GDG) agreed that health-related quality of life, pain severity, function, and psychological distress were the outcomes that were critical for decision making. Responder criteria (greater than 30% improvement in pain or function), adverse events, and healthcare utilization were also considered as important. Evidence was reported for all the outcomes except for psychological distress and healthcare utilization. For image-guided facet joint injections, evidence was only available for pain, function, and responder criteria. There was no evidence for any of the other outcomes.” The NICE summary concludes: “Overall, the GDG agreed that there was no consistent good quality evidence to recommend the use of spinal injections for the management of low back pain. There was minimal evidence of benefit from injections, and reason to believe that there was a risk of harm, even if rare. The GDG consequently agreed that it was appropriate to recommend against the use of spinal injections for people with low back pain.”
- The American Association of Neurological Surgeons (AANS) and the Congress of Neurological Surgeons (CNS) 2014 Guideline update for the performance of fusion procedures for degenerative disease of the lumbar spine: Part 13: injection therapies, low-back pain, and lumbar fusion. Lumbar IA facet injections are not recommended for the treatment of chronic lower-back pain. The literature does suggest the use of lumbar medial nerve blocks for short-term relief of facet-mediated chronic lower-back pain without radiculopathy.[42]
- NASS Evidence-Based Clinical Guidelines for Multidisciplinary Spine Care: Diagnosis and Treatment of Degenerative Lumbar Spondylolisthesis 2nd Edition (2014) reports that there is insufficient evidence to make a recommendation for or against the use of injections for the treatment of degenerative lumbar spondylolisthesis. Grade of recommendation: I (Insufficient Evidence).[11] The 2020 NASS Low Back Pain Guidelines states there is insufficient evidence to make a recommendation for or against the use of steroid injections for suspected facet-mediated pain (grade 1 recommendation) nor sufficient evidence for or against the use of 50% reduction in pain following MBB for diagnosis of facet joint pain (grade 1 recommendation). The guidelines also state

that the outcomes of RFA become more reliable when more stringent diagnostic criteria are used, and relief is durable for at least 6 months following procedure (grade B recommendation).[6] NASS Guidelines state while there is some evidence in the literature when compared to RFA, RFA offers longer-term pain relief, and long-term outcomes have not been reported in adequately designed trials. They state that if therapeutic IA injections are used, they should be used in patients whose initial injections resulted in pain relief >50% for at least 3 months and not be repeated more than 3 times annually. The guidelines address concern of the majority of research on therapeutic MBBs comes from a single-center, reports on prospective studies and 1 RCT which had unspecified intervals and cannot determine the true duration of treatment and concern with lack of studies comparing therapeutic MBBs to MB RFA.[6,11]

- Agency for Healthcare Research and Quality (AHRQ) Technology Assessment Program: Pain Management Injection Therapies for Low Back Pain (2015) authors used predefined criteria and selected randomized trials of patients with lumbosacral radiculopathy, spinal stenosis, nonradicular back pain, or chronic postsurgical back pain that compared effectiveness or harms of the epidural, facet joint, or sacroiliac corticosteroid injections versus placebo or other interventions. Also included were randomized trials that compared different injection techniques and large (sample sizes greater than 1000) observational studies of back injections that reported harms. Seventy-eight randomized trials of epidural injections, 13 trials of facet joint injections, and 1 trial of sacroiliac injections were included. Limited evidence suggested that epidural corticosteroid injections are not effective for spinal stenosis or nonradicular back pain and that facet joint corticosteroid injections are not effective for presumed facet joint pain.[39]
- A 2009 American Pain Society guideline recommends against the use of corticosteroids into the facet joint.[40]

Some societal recommendations that support the use of therapeutic facet join injections include:

- 2020 ASIPP Guidelines[8]: Therapeutic Facet Joint Interventions in Lumbar Spine states:
  - The level of evidence is II with moderate strength of recommendation for therapeutic lumbar facet joint nerve blocks with the inclusion of 3 relevant RCTs with long-term improvement.
  - The level of evidence is IV with a weak strength of recommendation for lumbar facet joint IA injections with inclusion of 9 relevant RCTs, with most of them showing lack of effectiveness without the use of local anesthetic.

The American Society of Anesthesiologists (ASA) Task Force on Chronic Pain Management and the American Society of Regional Anesthesia and Pain Medicine (ASRA) Practice Guidelines for Chronic Pain Management (2010)[41]:

- Recommendations for joint blocks: RCTs report equivocal findings regarding the efficacy of facet joint steroid injections compared with facet saline injections regarding pain relief for patients with low back pain. However, studies with observational findings for facet joint injections indicate that pain scores are improved over baseline scores for assessment periods of 1-6 months. IA facet joint injections may be used for symptomatic relief of facet-mediated pain. MBBs may be used for the treatment of facet-mediated spine pain.
  AANS and CNS 2014 Guideline update for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 13: injection therapies, low-back pain, and lumbar fusion states lumbar IA facet injections are not recommended for the treatment of chronic lower-back pain. The literature does suggest the use of lumbar

medial nerve blocks for short-term relief of facet-mediated chronic lower-back pain without radiculopathy.[42]

Radiofrequency Ablation

Radiofrequency neurotomy is described as radiofrequency lesioning performed utilizing either a heat lesion or pulsed mode radiofrequency. A thermal radiofrequency neurotomy lesion for facet denervation is performed at 80° to 85°C. Clinically, a higher temperature allows for a larger lesion to be made. The size of the lesion is influenced by the vascularity of the surrounding tissue: the greater the vascularity of the tissue, the smaller the lesion. Overall, the mechanism of radiofrequency neurotomy is described as denaturing of the nerves. Consequently, with radiofrequency, the pain returns when the axons regenerate, requiring repetition of the radiofrequency procedure. The pulsed mode radiofrequency is an application of a strong electric field to the tissue that surrounds the electrode, and the temperature of the tissue surrounding the tip of the electrode does not exceed 42°C and heat is dissipated during the silent period.[8]

The effectiveness of RFA to improve pain and function for facet joint mediated back pain in carefully selected individuals has been explored in the medical literature. Clinical trials report conflicting data on the effectiveness of RFA. Most clinical trials defined success with RFA as 50% pain relief at 6 months.

A 2015 Cochrane review by Maas, et al., including 23 RCTs (n=1309), evaluated patients who had a positive response to diagnostic block and underwent RFA found moderate evidence on the effectiveness of RFA compared to placebo for pain management of facet origin over the short term (mean difference (MD) -1.47, 95% confidence interval (CI) -2.28 to -0.67). Low-quality evidence that facet joint RFA is more effective than placebo for function over the short term (MD -5.3, 95% CI -8.66 to -0.20) and over the long term (MD -3.9, 95% CI -6.94 to 0.47). RFA for disc pain did not show effects compared to placebo over the short or long term (MD -1.63, 95% CI -2.58 to -0.68). They concluded no high-quality evidence to suggest RFA procedure provides pain relief for patients with chronic low back pain.[43]

Juch, et al. reported on 125 patients in the treatment arm and 126 in the control arm (exercise). Patients had a single diagnostic block with at least 50% or more pain reduction before RFA. The MD for the primary outcome pain intensity at 3 months was –0.18 (95% CI, –0.76 to 0.40). The MD for functional status at 3 months was –2.45 (95% CI, –5.53 to 1.03); the relative risk (RR) for global perceived recovery at 3 months was 1.35 (95% CI, 0.81 to 2.05). This study has been challenged, stating that the lack of efficacy was due to the treatment arm using the perpendicular technique.[44] Perpendicular electrode placement has been questioned with concern it may fail to reach the target nerve or only capture a segment, while parallel electrode placement reliably captures the target nerve and does so along a substantial length of the nerve. Those who challenge these studies report there are 3 studies (1999-2008) that utilized the parallel technique that would result in positive

outcomes and report support for RFA from these trials.[44] There are no recent trials that address this controversy.

Leggett, et al. (2014) conducted a systematic review to determine the efficacy of RFA for chronic low back pain associated with lumbar facet joints, sacroiliac joints, discogenic low back pain, and the coccyx. Included articles were sham-controlled RCTs, assessed the efficacy of RFA, reported at least 1 month of follow-up, and included participants who had experienced back pain for at least 3 months. Eleven sham controlled RCTs were included: 3 studies involving discogenic back pain, 6 studies involving lumbar facet joint pain, and 2 studies involving sacroiliac joint pain. No studies were identified assessing the coccyx. The evidence supports RFA as an efficacious treatment for lumbar facet joint and sacroiliac joint pain, with 5 of 6 and both of the RCTs demonstrating statistically significant pain reductions, respectively. The evidence supporting the RFA for the treatment of discogenic pain is mixed. Future studies should examine the clinical significance of the achieved pain reduction and the long-term efficacy of RFA.[45]

The systematic review (2015) by Manchikanti, et al. looked at both therapeutic injections, and RFA in lumbar, cervical and thoracic regions evaluated 21 RCTs and 5 observational trials, using a modified approach to the grading of evidence[4], reported level II evidence[3] for RFA in the lumbar and cervical spine, and level IV in the thoracic region.[26] A systematic review by Engel, et al. (2016), which serves as the basis for NASS recommendations, “The Effectiveness and Risks of Fluoroscopically-Guided Cervical Medial Branch Thermal Radiofrequency Neurotomy: A Systematic Review with Comprehensive Analysis of the Published Data” using the GRADE system for rating. They conclude the majority of patients were pain-free at 6 months and over a third pain-free at 1 year.[27] In 2017, Lee, et al. conducted a meta-analysis that included 7 trials with 454 patients comparing RFA (n=231) to sham or epidural (n=223) and reported greater improvement in the RFA group.[46]

2020 ASIPP Consensus Guidelines[8] Table 22 reports on 11 studies on RFA effectiveness and concludes:

Lumbar Spine: The level of evidence is II with moderate strength of recommendation for lumbar RFA with the inclusion of 11 relevant RCTs with 2 negative studies and 4 studies with long-term improvement.

ASA Task Force on Chronic Pain Management and ASRA Practice Guidelines for Chronic Pain Management (2010) Recommendations for Ablative Techniques[41]: The Task Force notes that other treatment modalities should be attempted before consideration of the use of ablative techniques.

- Chemical denervation: Chemical denervation (e.g., alcohol, phenol, or high-concentration local anesthetics) should not be used in the routine care of patients with chronic non-cancer pain.

- Cryoablation: Cryoablation may be used in the care of selected patients (e.g., post-thoracotomy pain syndrome, low back pain (MB), and peripheral nerve pain).
- RFA: Conventional (e.g., 80°C) or thermal (e.g., 67°C) RFA of the MB nerves to the facet joint should be performed for low back (MB) pain when previous diagnostic or therapeutic injections of the joint or MB nerve have provided temporary relief. Conventional RFA may be performed for neck pain. Conventional or thermal RFA of the dorsal root ganglion should not be routinely used for the treatment of lumbar radicular pain.

NICE guideline on low back pain and sciatica in over 16s: assessment and management (2016) evidence statement[47]:

- Radiofrequency denervation compared with placebo/sham for low back pain: Evidence from 4 studies demonstrated clinical benefit in pain for radiofrequency denervation compared to placebo/sham at both the short- and long-term follow-ups of less than and greater than 4 months (low to moderate quality, n=160). In contrast, there was no difference in function between treatments at any time point. Conflicting evidence from 1 study for quality of life at less than 4 months follow-up showed clinical benefit for radiofrequency denervation compared to placebo/sham for the Short Form 36-Health Survey (SF-36) domains of general health and vitality. Radiofrequency denervation was inferior to sham for the domains of mental health, pain, and social function. There was no difference between treatments for the physical domain (low quality, n=81).
  Evidence from a single study reporting adverse events at less than 4 months follow-up demonstrated an increase in adverse effects for radiofrequency denervation in terms of the number of patients with moderate or severe treatment-related pain (low quality, n=79). There was no difference in other adverse events (change of sensibility and loss of motor function) at short-term follow-up when radiofrequency denervation was compared to placebo/sham in the same study (very low quality). Additionally, when compared with placebo/sham, a benefit for radiofrequency denervation in responders to pain reduction measured by global perceived effect was demonstrated by 2 studies at both the less than and greater than 4 months follow-up time points. However, this was not seen for pain reduction measured by VAS at less than 4 months reported by a single study (low quality, n=111).

- Radiofrequency denervation versus MBB: Evidence from a single study demonstrated clinical benefit in terms of pain for radiofrequency denervation compared to MBBs at both the short- and long-term follow-ups of less than and greater than 4 months (very low quality, n=100). Radiofrequency denervation has evolved as a treatment for spinal pain over the last 40 years and is a minimally invasive and percutaneous procedure performed under local anesthesia or light intravenous sedation. Radiofrequency energy is delivered along an insulated needle in contact with the target nerves. This focused electrical energy heats and denatures the nerve. This process may allow axons to regenerate with time, requiring the repetition of the radiofrequency procedure.
- The duration of pain relief following radiofrequency denervation is uncertain. Data from RCTs suggests relief is maintained for at least 6-12 months, but no study has reported longer-term outcomes. Pain relief for more than 2 years would not be an unreasonable clinical expectation.

- The de novo economic model undertaken for this guideline for radiofrequency denervation suggested that the treatment is likely to be cost-effective, provided the duration of effect exceeds 16 months.
- If radiofrequency denervation is repeated, we do not know whether the outcomes and duration of these outcomes are like the initial treatment. If repeated radiofrequency denervation is to be offered, we need to be more certain that this intervention is both effective and cost effective.

NASS Guidelines state that therapeutic MB RFA is a validated treatment for facet mediated pain, and repeat procedures are equally successful if the response to the initial RFA lasted at least 3 months. These guidelines advocate dual diagnostic MBB with ≥80% relief of the primary (index pain), and the onset, and minimum duration of relief is consistent with the agent employed. Also, RFA should be performed at the same level no more than twice annually, and only if the initial radiofrequency lesion results in significant pain relief (> 50%) for at least 6 months. In those situations, a repeat procedure in that year is appropriate.[14] 2020 NASS Low Back Pain Guidelines states there is insufficient evidence to make recommendations for or against the use of cryodenervation for treatment of zygapophyseal (facet) joint pain.[6]

AANS and CNS 2014 Guideline update for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 13: injection therapies, low-back pain, and lumbar fusion. Lumbar IA facet injections are not recommended for the treatment of chronic lower-back pain. The literature does suggest the use of lumbar medial nerve blocks for short-term relief of facet-mediated chronic lower-back pain without radiculopathy. Lumbar medial nerve ablation is suggested for 3–6 months of relief for chronic lower-back pain without radiculopathy. Diagnostic medial nerve blocks by the double-injection technique with an 80% improvement threshold are an option to predict a favorable response to medial nerve ablation for facet-mediated chronic lower-back pain without radiculopathy. Still there is no evidence to support the use of diagnostic medial nerve blocks to predict the outcomes in these same patients with lumbar fusion.[42]

Cervical and Thoracic Spine

Most of the literature focuses on the lumbar region regarding interventional facet joint diagnosis, and the 2020 Consensus Guidelines are specific for the lumbar region. The 2020 ASIPP Guidelines include the cervical and thoracic regions. The 2020 ASIPP Guidelines report level I-II with moderate to strong strength of recommendation for lumbar, diagnostic blocks report 10 diagnostic accuracy studies to support the recommendation. ASIPP report level II evidence for the cervical spine with moderate strength or recommendation citing 10 diagnostic accuracy studies. Two systematic reviews support reliance on dual diagnostic blocks with 100% relief of the index pain to select patients for the procedure reporting more than 60% selected using this stringent criterion had 100% relief of index pain. They were

pain-free at 6 months and nearly 40% pain free at 1 year with functional improvement in neck pain.[27,28]

ASIPP (2020)[8] Therapeutic Facet Joint Intervention Cervical Spine:

- The level of evidence is II with moderate strength of recommendation for therapeutic cervical facet joint nerve blocks with the inclusion of 1 relevant RCT and 3 observational studies, with long-term improvement.
- The level of evidence is V with weak strength of recommendation for cervical IA facet joint injections with the inclusion of 3 relevant RCTs, with 2 observational studies, the majority showing lack of effectiveness. In contrast, 1 study with a 6-month follow-up showed a lack of long-term improvement.
- The level of evidence is II with moderate strength of recommendation for cervical RFA with the inclusion of 1 RCT with positive results and 2 observational studies with long-term improvement.

Evidence to support use in thoracic region is further supported by one 2012 systematic review report evidence for diagnostic accuracy of the thoracic facet joint injection, and one 2018 study comparing thoracic IA vs. MBB reporting significant pain relief with thoracic joint injection.[48] NASS guidelines state the lack of supporting literature for the use of IA injections for thoracic pain, however, states the lack of other effective alternative therapies.[11]

Imaging

Image guidance has become standard care for facet blocks and RFA procedures, allowing accurate needle placement, monitor contrast and anesthetic spread, and improve safety by direct visualization of bony elements and avoiding surrounding structure. Studies dating back to the 1980s comparing “blind” injections to image-guided concluded that the procedure should not be performed without radiographical imaging. Fluoroscopy is considered the gold standard for facet block procedures, and most studies were conducted with fluoroscopic guidance and consistent with societal recommendations. The U.S. Department of Health and Human Services Office of Inspector General (OIG) report in 2008 asserted radiographic guidance is recommended.[49] One cadaveric study showed improved accuracy with fluoroscopy over CT.[50] 2020 Consensus Guidelines recommend CT or preferably fluoroscopy for lumbar MBB (grade B recommendation), CT scanning for IA placement (grade C recommendation), and fluoroscopy for MB RFA (grade B recommendation, low level of evidence). The guidelines state fluoroscopy is preferred to CT due to lower cost, faster times, and less radiation exposure than CT.[5]

The use of ultrasound has been explored as an alternative imaging modality. Cadaveric, retrospective studies, and review of RCT and non-RCT suggest that there is little difference between ultrasound and fluoroscopy guided procedures; however, there are no well-designed studies to report the safety of this approach. Limitations for ultrasound guidance include reduced visibility in obese patients, more challenging to target the proper segment, and not being able to visualize facet joint.[5] NASS Guidelines state insufficient evidence to

support use of ultrasound guidance.[11] SMEs score low confidence that the facet joint procedures can be performed under ultrasound guidance (score 1.7/5).

Facet Joint Cyst Rupture

Facet joint cysts have been found to have a prevalence of 6.5% and are thought to play a role in facet mediated pain. In these cases, nerve root compression or irritation is associated with radicular pain like other neuro-compressive lesions. Rupture of the cyst has been associated with pain improvement and may potentially avoid more invasive open surgical procedures. In these cases, IA injection is used and often with TFESI to treat radicular pain.[11] A systematic review of 870 patients report cyst resolution to be 58% for the percutaneous procedure, and 90% for decompressive procedures concluding advantage of surgical intervention as compared to percutaneous procedures and pain relief with surgical management is estimated 83.5%. Shuang, et al. systematic review of 29 studies reported overall the satisfactory results (after short- or long-term follow-up) were achieved in 55.8% [49.5, 62.08] (pooled mean and 95% CI) of the 544 patients subjected to percutaneous lumbar facet joint cyst resolution procedures. 38.67% [33.3, 43.95] of this population underwent surgery subsequently to achieve durable relief.[51] However, the analysis consisted largely of small retrospective studies and case reports with low-quality evidence. In 1 prospective, non-randomized study, 120 patients' satisfaction was reported as 75%, with 25% requiring repeat surgery. NASS Guidelines address the concern of cyst reoccurrence, which occurs about 50% of the time and does not recommend repeating the procedure more than once and only if the first procure produces satisfactory results, which they define as 50% improvement for at least 3 months.[11] The SME panel voted 3.9/5 in support of facet joint cyst rupture for pain relief.

Centralized Pain Syndrome

Facet procedures by nature treat a local pain condition, while a centralized pain syndrome, such as fibromyalgia, is a generalized pain condition. Patients with centralized pain syndromes are thought to have alternations in central neurotransmission. A prospective study with 548 patients diagnosed with primary spine pain were also evaluated for fibromyalgia criteria before spinal interventions. 42% of the patients meet diagnostic criteria for fibromyalgia. They determined a profound phenotypic difference in those with fibromyalgia as compared to those with spinal pain alone associated with more neuropathic pain descriptors, anxiety, greater pain interference and lower physical function. The authors concluded that the phenotypical factors found in the fibromyalgia group may be predictive of poor outcomes in spine intervention procedures.[52] Exploration of factors associated with poor outcomes from lumbar RFA and long duration pain was correlated with treatment failure.[53] There is insufficient literature that addresses if facet joint procedures are beneficial for centralized pain conditions.

Intrafacet Implants

Intrafacet implants have been proposed as an alternative technique to surgical fusion. They involve the placement of an allograft dowel made from bone (from femur or tibia) and placed surgically or via the minimally invasive procedure. The allograft, which is processed by licensed tissue banks, which must be compliant with FDA requirements for tissue processing, are not subject to FDA 510K clearance and can be marketed. There are no clinical trials that address the efficacy and safety of these implants. The report on 6 cases did not indicate efficacy.[54]

Non-Thermal Modalities

Several alternatives to percutaneous radiofrequency denervation have been proposed, including pulsed radiofrequency, cryoablation, laser ablation, and chemical ablation, in which a neurolytic substance (e.g., alcohol, phenol, glycerol) is injected into the affected nerve root. An alternative method of denervation using an endoscopic approach (i.e., endoscopic dorsal ramus rhizotomy) has also been proposed.[55] The literature on these modalities is largely limited to case reports or small series with mixed results.[14,56-58] One RCT (n=80) comparing pulsed RFA to steroid injections concluded it may be more effective than steroids.[59] A RCT (n=50) compared study of continuous to pulsed RFA and did not report significant differences.[60] Long-term safety and efficacy data is lacking. The ASA Task Force on Chronic Pain Management and the ASRA Practice Guidelines for Chronic Pain Management (2010) Recommendations for Ablative Techniques: The Task Force notes that other treatment modalities should be attempted before consideration of the use of ablative techniques.[41]

Anesthesia

ASA provides definitions on the continuum of the depth of sedation. Minimal sedation anxiolysis is defined as a drug-induced state during which patients respond normally to verbal commands. Although cognitive function and physical coordination may be impaired, airway reflexes, and ventilatory and cardiovascular functions are unaffected. Progressing depth of sedation beyond minimal is moderate or “conscious”, deep and general. MAC is not a depth of sedation, but a specific anesthetic service allowing a deeper level of analgesia than can be provided by moderate sedation. ? Statement on ?Continuum of Depth of Sedation: Definition of General Anesthesia and Levels of Sedation/Analgesia.[61]

The increased risk associated with the use of anesthesia, including sedation, in the elderly population must be considered. A 2020 review reports a substantial increase in the use of non-operating room anesthesia (NORA) for elderly patients and that advanced age is an independent risk factor for anesthesia-related adverse events in this population. The authors explain there are significant changes in pharmacological effects of anesthetic agents with

aging requiring reduced dosing and careful titration and monitoring to reduce adverse events. They also report that increasing age increases the risk of aspiration and cognitive complications even during sedation. The paper discusses that older and frailer patients may benefit greatly from less invasive procedures, but this must be balanced against safety outside of a high acuity operating suite and increased risk associated with anesthetic in this population.[62] A retrospective analysis from the 2008-2012 Premier Database reviewed 21,276,691 inpatient discharges seeking a better understanding of the risk of cardiopulmonary resuscitation (CPRA). They found 96,554 patients suffered CPRA and reported patients who received opioids and sedatives had an adjusted odds ratio for CPRA of 3.47 (95% CI: 3.40-3.54; p<0.0001) compared with patients not receiving opioids or sedatives. Therefore, opioids alone and sedatives alone were associated with a 1.81-fold and a 1.82-fold (p<0.0001 for both) increase in the odds of CPRA, respectively. They found 42% survived CPRA and only 22% were discharged to home calling for further research in this area and efforts to investigate opioid sparing anesthesia and reduced sedative use and improved monitoring given this risk.[63]

The ASA committee opinion for anesthetic care during interventional pain procedures for adults stating that when sedation is provided during the performance of pain procedures it is important that the patient can be responsive during critical portions of the procedure to report potential procedure related paresthesia, acute changes in pain intensity or function for potential toxicity. The committee opinion states that interventional pain procedures generally only require local anesthetic, however, patients may elect to also receive supplemental sedation but must remain conscious. Exceptions may include a well-documented history of inability to cooperate, medical conditions that would prohibit performance of the procedure, or inability to remain motionless.[1]

The Spine Intervention Society Factfinder for patient states that sedation is not intrinsically necessary for interventional spine procedures and that the decision should be considered on case-by-case basis.[64] This report focused on moderate sedation and expresses concern of over-sedation impairing the ability “of the patients to respond adequately to any potential pain, discomfort, or paresthesia during a procedure and warn the physician accordingly of a potential impending serious complication.” The use of heavy sedation has been associated with increased risk of neurological injuries, such as spinal cord injury. Moderate sedation has been associated with apnea and hypoxemia. While some reports common use of sedation, there is limited evidence of isolated situations where sedation may provide medical benefit. A RCT with 73 subjects with crossover design reported sedation did not predict satisfaction with care or improved pain.[65] A report of 6897 patients with TFESI without sedation reported only a 3% dissatisfaction rate.[66] The reports concluded the default use of sedation is not associated with clear health/outcome benefits and associated with rare but serious increased risk. That most patients are satisfied with care without sedation and that it should

be used judiciously and under circumstances with careful monitoring and administered by trained providers and with appropriate monitoring.

Reconsideration Request

A reconsideration request was received requesting therapeutic facet joint injections be considered as a first line treatment option for facet pain and include both IA injections or MBBs. Most of the submitted literature was already in the evidence section of the LCD. Additional submitted studies published prior to 2016 were not cited individually in the LCD and many were included in at least 1 of the systematic reviews that were evaluated in the LCD.[26,37] Literature evaluated with this request include:

- A 2022 retrospective report comparing 99 patients who received lumbar facet joint nerve blocks compared to 227 receiving radiofrequency neurotomy with 1-year follow-up. Pain improvement was reported in both groups with 100%, 99% and 79% reporting improvement in the nerve block group as compared to 100%, 74% and 65% in the RFA group at 3, 6 and 12 months respectively. Patients received between 1-4 nerve blocks and 1-2 RFAs. The authors conclude that the outcome from therapeutic facet joint nerve blocks is comparable to RFA. Missing data with 48 of the original 99 and 148 of the original 227 reported at 12 months data limits this data set.[67] A similar study was conducted to evaluate 132 patients who received cervical MBBs and 163 patients receiving cervical RFA. Of the original group, 107 in the injection group and 105 in the RFA group completed 12-month follow-up and reported 100%, 94%, and 81% of the patients in the MBBs group, and 100%, 69%, and 64% in the radiofrequency neurotomy group at 3-, 6-, and 12-month follow-up, Both studies are limited by the retrospective design and without a control group cannot determine if the procedures are equivalent.[68]
- A European systematic review explores the management of facet joint osteoarthritis with MBBs. Eight articles were included in the review. The studies were challenged by small sample sizes ranging from 8 to 60 and the majority having less than 30 subjects, short- term follow-up with a maximum of 6 months, significant variability in treatments, different tools to measure pain and other limitations leading the authors to conclude that the role of facet injection in the management of low back pain has a positive trend but is non-conclusive.[69]
- A small RCT compared 60 subjects with a single positive diagnostic block receiving 1-2 level treatment with IA injection or RFA for 6 months. They reported a decrease in NRS score at 2 weeks, and 1,3, and 6 months for both groups. They report similar outcomes at 6 months for both groups. The authors reported steroid related side effects in the injection group and no adverse events in the RFA group suggesting that the RFA may be a better option for management to reduce the potential adverse events related to steroids.[70]
- A small RCT with 40 subjects randomized patients with cervical facet joint pain to IA injection or RFA. They report both groups had a decrease in NRS at 1, 3 and 6 months (p=0.000). They report both groups reported ≥50% and the author concludes that RFA is an effective treatment for cervical facet pain while avoiding the adverse effects of steroids.[71]
- A small prospective study administered a sham block followed by MBB with presumed facet mediated pain to 104 subjects. Patients who experienced improvement with the saline solution were excluded from the study for presumed placebo effect (16.7%) and patients who did not experience at least 50% improvement after the lidocaine injection were referred for evaluation for other potential

etiologies (31.7%). The remaining 54 patients were followed with VAS score measurements and 67% reported improvement maintained at 3 months. The remaining 33% reported worsening pain scores over this duration. The authors concluded that this group was the true facet mediated pain while the group that experienced improvement after a single diagnostic MBB had myofascial pain which resolved during this 3-month follow-up. This is consistent with data that suggests a high false-positive rate with a single MBB and supports the use of dual diagnostic blocks and encourages 3-month follow-up injections to ensure correct diagnosis.[72]

- A 2023 systematic review and meta-analysis compiles 11 studies (3 RCT, 8 observational) on MBB and RFA for thoracic pain. The authors acknowledge the paucity of high-quality literature and conclude Level II evidence for MBB and Level III evidence for RFA for long-term management of chronic thoracic pain.[73] Like the above systematic review, this GRADE recommendations are limited by the use of risk of bias tool unique to interventional pain.
- A 2023 RCT compared IA to MBB for facet injections for 60 patients with lumbar facet joint pain. Thirty patients received IA blocks vs 30 patients with MBB, and it was found there was statically significant improvement ($p>0.05$) in both groups.[74] The authors conclude both modalities are safe and effective for administering facet injections. Another 2023 RCT includes 60 patients with chronic facet pain and were randomized to MBB with lidocaine (n=31) or bupivacaine (n=31) and found no difference between the groups for 16 weeks.[75] Both studies were limited by small size and short-term follow-up.
- A 2024 systematic review was conducted to evaluate the effectiveness of facet nerve blocks as a therapeutic modality for facet joint pain. Risk of bias assessment was assessed with the Interventional Pain Management Techniques – Quality Appraisal of Reliability and Risk of Bias Assessment (IPM-QRB) for randomized therapeutic trials, and the Interventional Pain Management Techniques – Quality Appraisal of Reliability and Risk of Bias Assessment for Nonrandomized Studies (IPM-QRBNR) for nonrandomized studies.[76] Recommendations were developed with GRADE assessment criteria. They found only 3/21 studies showed high-level of evidence and 11 studies with moderate-level of evidence concluding Level II Evidence to support therapeutic joint nerve blocks for facet mediated spinal pain. The GRADE conclusion is limited by risk of bias assessment being conducted with a tool unique to interventional pain management rather than the risk of bias tools typically used as part of the GRADE analysis.

The new evidence to support the role of therapeutic injections as first line therapy was very low-quality. Additional literature supports the coverage position of RFA if dual diagnostics blocks are positive for facet mediated pain. Therefore, there was not sufficient evidence submitted with this reconsideration to change the LCD's coverage position and allow therapeutic injections as first line treatment for facet mediated pain. However, evidence that the effectiveness and outcomes for IA and MBB are similar and, therefore, either should be allowed when therapeutic RFAs are appropriate is supported by evidence.

Analysis of Evidence (Rationale for Determination)

Facet joint procedures are a challenging area due to variability in the literature, lack of consensus among experts, differences in societal guidelines and a historical pattern that demonstrates high risk for over utilization. When there is a lack of consensus on best practices, careful evaluation of the medical literature and utilization of the best available

evidence serves as the basis for our determinations in coverage and guidelines. This is supplemented with the knowledge shared from our SME panel.

The literature and consensus of the expert panel and societal guidelines agree that there is no specific history, physical exam findings, or imaging studies that can diagnose facet joint mediated pain and predict response to treatment. For this reason, diagnostic facet joint injections are indicated. MBBs are the preferred technique for these injections, which is validated and supported by literature as compared to the IA injections, which lack validation, are technically more difficult, and more painful. There is consensus that all facet procedures should be performed under image guidance, CT or fluoroscopy, with a preference for fluoroscopy.

The controversial areas are the number of blocks and the percentage of pain relief required to consider a block successful. For the diagnostic block, the pain relief achieved is temporary and used as a predictor of success for subsequent RFA. The objective is to reduce false-positives that would not be predictive of success with RFA and avoid false-negative patients who could benefit but would not be offered treatment. Many studies have used a cut-off of pain relief greater than 75-80% to consider a diagnostic block as positive. Ten studies assessing the prevalence of lumbar facet joint pain using a cut-off value of 80% relief reported with a prevalence rate of 27% to 40% with false-positive rates of 27% to 47% received a moderate to strong recommendation from 2020 ASIPP Guidelines.[8]

The 2020 Consensus Guidelines advocate for using single blocks with ≥50% for clinical trials and clinical practice while acknowledging that the existing evidence does not adequately address the 50-80% group and that this cut-off was selected to maximize access to care given lack of reliable alternative treatment options in this population and potential benefit of treatment in this group.[5] On the contrary, there are studies that show patients with dual blocks using ≥80% relief are more likely to show a positive response to RFA. This aligns with ASIPP and NASS guidelines.[2,11] Given the lack of clear evidence, the consensus among experts and societies is that there is uncertainty regarding the cut-off values and number of blocks. Most of the evidence and societal guidance supports the use of a more stringent diagnostic cut-off. The multi-MAC SME panel was a score of 3.2/5 favoring the dual block regimen. Once a patient begins treatment, the treatments often continue without repeat diagnostic studies, so the correct diagnosis initially is important. The existing evidence is clear there is a high risk of false-positives with a single block and 50% cut-off and that a dual block and higher cut-off (75-80%) improves diagnostic accuracy. The evidence also supports that improved diagnostic accuracy predicts greater improvement with RFA treatments. While this must be balanced against excluding some patients for treatment who may potentially benefit, given the lack of supporting literature to define this population, current evidence supports dual blocks at 75-80% improvement is more strongly supported than the alternative less stringent approach.

The effectiveness and clinical utility of therapeutic facet blocks is another controversial area. In this case, we are referring to injections with corticosteroids as injections as local anesthetic would not be expected to have any sustained effect. There is support for the use of therapeutic injections in the form of a systematic review and societal guidelines. Most of this support is from a single center that utilizes a published, modified grading criterion they developed. Grading from 1 grading criteria to another often have similarities but cannot be extrapolated to another so we must consider each grading criteria individually. There is some additional literature from other sites and societal support for therapeutic injections offered by ASA and ASRA. The overwhelming majority of other guidelines and evidence do not support therapeutic IA injections, and this includes multiple studies, systematic reviews, and guidelines by 2020 Consensus Guidelines, NASS, NICE, AHRQ, AANS, and CNS. Their use received a “D” rating from 2020 Consensus Guidelines. The current consensus is that diagnostic injections are to select patients who may be candidates for RFA, and not for therapeutic injections. Clinically since the reported efficacy of therapeutic injections (if a true benefit exists) would be around 3 months compared to 6 months or longer for RFA, so there is an obvious benefit for selection of RFA as a treatment over injections. The multi-MAC SME voting was 3.4/5 that subsequent therapeutic IA or MBB are effective. The evidence does support several special circumstances when RFA cannot be performed, and therapeutic injections may be appropriate. Therefore, the policy allows therapeutic injections in circumstances when RFA is not feasible.

The goal procedure for facet joint interventions is RFA. The strongest available data related to facet interventions is to support the use of RFA for facet joint pain. Still, even that is mixed and lacks a clear consensus on proper patient selection and effectiveness. A 2015 Cochrane review showed a moderate benefit for RFA for pain, but lack of benefit for function.[43] However, the patient selection used the less stringent criteria of a single block with a 50% reduction in pain. Other studies have shown this criterion is associated with a higher false-positive rate and less success with RFA procedures, so this conclusion may not be represented if the stricter criteria were used. The FACTS study, an RCT that compared outcomes stratified by different prognostic values and reported the success rate in those who underwent RFA in the zero (no block), single and double block groups were 33%, 39%, and 64%, respectively.[31] While this served as the basis for recommending a single block at 50% in the 2020 Consensus Guidelines, it also demonstrates the improved success rate for the dual block group, which increased to 64%.[19] A systematic review that used the GRADE system for rating which is a more standardized grading system and allows both RCT and observational studies to be evaluated, concluded the majority of patients were pain free at 6 months and over a third pain free at 1 year.[27] This conclusion was supported by a 2017 meta-analysis.[46] The overall societal consensus is in support of the use of RFA with ASIPP, ASA/ASRA, NICE, 2020 Consensus Guidelines, and AANS/CNS, all having guidelines that support RFA use. Even the NASS Guidelines that report insufficient evidence for RFA provided guidelines for use. There remain concerns as the clinical studies of RFA for chronic low back pain have significant methodological limitations that can affect the interpretation of the data. Uncertainties regarding several aspects of RFA for spinal pain, necessitate additional research. The validation of radiofrequency for chronic spinal pain management

relies upon the resolution of these technical issues, as well as issues regarding patient selection and long-term efficacy.[45] While the literature is not complete in terms of efficacy and patient selection for RFA facet procedures, the procedure offers the potential to avoid more invasive surgery. It may potentially reduce opioid use, justifying maintaining RFA as an option in the management of back pain in properly selected patients despite these limitations.

Limitations, such as number of levels and frequency, are challenging as there is little evidence focusing on these factors. Lack of consensus in the medical community about frequency of injections has been a challenge.[49] Societies have begun to offer recommendations on frequency. The NASS Coverage Recommendations recommends dual blocks, does not recommend therapeutic procedures, recommends limiting RFA to maximum of twice per year and procedures for facet joint cyst rupture to be repeated not more than once.[11] The 2020 Consensus Guidelines report recommends single diagnostic blocks, does not recommend therapeutic injections and recommends limiting RFA to no more than 2 times per year.[5] The multi-MAC SME panel also provided input on frequency. The panel voted 1.8/5 that diagnostic injections should be a minimum of 28 days apart and recommended allowing sooner may be beneficial. The panel voted 3.5/5 that therapeutic injections should be a minimum of 3 months apart, and during treatment phase interventional procedures should be repeated only if medically necessary, and not to exceed 4 times per year was scored as 3.4/5. The panel voted RFA should be repeated only if medically necessary at a minimum of 6 months apart with a score of 3.9/5. The panel supported allowing 3 levels (diagnostic or therapeutic) in 1 session with score of 3.8/5; however, prevalence data demonstrates that 3 diagnostic levels are not routine and would be under unusual circumstances. There is currently no evidence to support that facet mediated pain may extend routinely beyond 2 levels. The purpose of the dual diagnostic block is to further identify the pain generator and successfully treat the impacted region. This correlates with literature when the number of levels is reported, Medicare national data and MAC local data, which demonstrates that most facet procedures involve 1 or 2 levels only.[77]

Several alternatives to percutaneous radiofrequency denervation have been proposed, including pulsed radiofrequency, cryoablation, laser ablation, and chemical ablation, in which a neurolytic substance (e.g., alcohol, phenol, glycerol) is injected into the affected nerve root. An alternative method of denervation using an endoscopic approach (i.e., endoscopic dorsal ramus rhizotomy) has also been proposed. There is insufficient evidence in the published medical literature to determine the safety and efficacy of these emerging alternative modalities or approaches compared to thermal radiofrequency denervation for the treatment of spinal pain.

General Information

Associated Information
N/A

Sources of Information
Reviewed but not cited:

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Revision History Information

Revision History Date	Revision History Number	Revision History Explanation	Reasons for Change
04/09/2026	R7	Under ***Coverage Indications, Limitations and/or Medical Necessity*** under **C. Facet Joint Denervation** revised b. to read "Repeat thermal facet joint RFA at the same anatomic site is considered medically reasonable and necessary provided the	• Provider Education/Guidance

Revision History Date	Revision History Number	Revision History Explanation	Reasons for Change
		patient had a minimum of consistent 50% improvement in pain for at least 6 months **or** at least 50% consistent improvement in the ability to perform previously painful movements and ADLs as compared to baseline measurement using the same scale."	
10/23/2025	R6	Under ***Coverage Indications, Limitations and/or Medical Necessity*** under section **C. Facet Joint Denervation**, under b., revised verbiage to "Repeat thermal facet joint RFA at the same anatomic site is considered medically reasonable and necessary provided the patient had a minimum of consistent 50% improvement in pain **or** consistent improvement in the ability to perform previously painful movements and ADLs as compared to baseline measurement using the same scale for at least 6 months." Under ***Bibliography*** updated the hyperlink for source #12.	• Provider Education/Guidance
07/17/2025	R5	Under ***Coverage Indications, Limitations and/or Medical Necessity*** under section C. Facet Joint Denervation, removed "ALL" from the sentence "The thermal radiofrequency destruction of	• Provider Education/Guidance

Revision History Date	Revision History Number	Revision History Explanation	Reasons for Change
		cervical, thoracic, or lumbar paravertebral facet joint (medial branch (MB)) nerves are considered medically reasonable and necessary for patients who meet ALL the following criteria." Added "Initial thermal radiofrequency destruction" to the beginning of (a.) and removed "AND" from the end of (a.). Under ***Bibliography*** removed broken hyperlink on reference #12 and changes were made to citations to reflect AMA citation guidelines.	
07/07/2024	R4	Under ***Coverage Indications, Limitations and/or Medical Necessity*** in section **B. Therapeutic Facet Joint Procedures (IA or MBB)** under Frequency Limitation removed "(IA)". Under **Limitations** removed "IA" in limitation #9.	• Provider Education/Guidance
07/07/2024	R3	This LCD is being presented for notice. Under ***Coverage Indications, Limitations and/or Medical Necessity*** in section B. title, "or MBB" was added. Under section D. Frequency Limitation, "per individual cyst" was added. Under Limitations #3, "anesthesia" was revised to "sedation" and the verbiage "or cyst rupture/aspiration" was added. Under Limitations #8, "or" was	• Provider Education/Guidance • Reconsideration Request

Revision History Date	Revision History Number	Revision History Explanation	Reasons for Change
		revised to “and”. Under ***Summary of Evidence*** under Reconsideration Request, the sixth, seventh, and eighth bullet points were added; and the following verbiage was added to the last paragraph, in the last sentence: “and allow therapeutic injections as first line treatment for facet mediated pain. However, evidence that the effectiveness and outcomes for IA and MBB are similar and, therefore, either should be allowed when therapeutic RFAs are appropriate is supported by evidence.” Under ***Bibliography*** sources 73-76 were added.	
05/18/2023	R2	Under ***CMS National Coverage Policy*** updated section headings and added the following regulations: “Title XVIII of the Social Security Act, §1861(s)(2) addresses medical and other health services; Title XVIII of the Social Security Act, §1862(a)(14) defines other than physician services; 42 CFR §410.74 defines physician assistants' services, §410.75 defines nurse practitioners' services, §410.76 defines clinical nurse specialists' services, and §419.22 defines hospital services excluded from payment under the hospital outpatient prospective payment system.” Under ***Bibliography*** fixed	• Provider Education/Guidance

Revision History Date	Revision History Number	Revision History Explanation	Reasons for Change
		hyperlink on reference #11, and changes were made to citations to reflect AMA citation guidelines. Formatting, punctuation and typographical errors were corrected throughout the LCD. Acronyms were inserted and defined where appropriate throughout the LCD.	
05/26/2022	R1	Under ***Summary of Evidence – Diagnostic Facet Joint Injections*** added reference #25 where appropriate. Under ***Bibliography*** repaired the broken hyperlinks for references #5 and #11.	• Provider Education/Guidance

Associated Documents

Attachments
N/A

Related Local Coverage Documents
Articles
A58350 - Billing and Coding: Facet Joint Interventions for Pain Management
A59751 - Response to Comments: Facet Joint Interventions for Pain Management
LCDs
DL38765 - Facet Joint Interventions for Pain Management (MCD Archive Site)

Related National Coverage Documents
NCDs
30.3 - Acupuncture
220.1 - Computed Tomography
150.7 - Prolotherapy, Joint Sclerotherapy, and Ligamentous Injections with Sclerosing Agents

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Keywords

- Facet Joint Interventions
- Pain Management